ENGINEERING DRAWING WITH AUTOCAD

UNIT 4: SECTIONAL VIEWS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 4.1 Concept and Need for Sectional Views

When an engineering component contains intricate internal cavities, bores, counterbores, and passageways, standard orthographic views become cluttered with confusing overlapping dashed hidden lines:
* Sectional View: An imaginary cutting plane cuts through the component, the portion between the observer and the plane is removed, revealing the interior cross-sectional geometry directly with solid visible lines.
* Cutting Plane: Imaginary slicing plane indicated by Cutting Plane Line (chain thin line with thick ends and direction arrows showing line of sight labeled A-A).
* Section Lining / Hatching: The solid material sliced by the cutting plane is highlighted with thin, continuous parallel hatching lines.

>> 4.2 Standard Hatching Rules and Conventions (BIS SP 46)

* Angle & Spacing: Drawn at 45 degrees to the main baseline of the view (or 30/60 deg if 45 deg parallels an object edge). Spacing uniform between 1.5 mm and 3.0 mm depending on view scale.
* Adjacent Components in Assembly: Hatched in opposite directions (e.g., 45 deg left vs 45 deg right) or with differing line pitches to clearly distinguish distinct mating parts.
* Non-Sectioned Standard Elements (Parts NOT Hatched along their axis):
- Shafts, bolts, nuts, studs, screws, rivets, keys, cotters, pins.
- Spokes and arms of wheels, gear teeth, ball and roller bearings.
- Thin webs, ribs, and stiffener plates (when cut longitudinally parallel to their plane, they are left unhatched to avoid giving a false impression of solid thickness).

>> 4.3 Classification of Sectional Views

* 1. Full Section: Cutting plane passes straight through the entire component along

its centerline (180 deg cut). Shows complete interior.
* 2. Half Section: Used for symmetrical objects. Cutting plane cuts halfway through (90 deg cut). One half is drawn in section showing interior, while the other half remains as an exterior view showing outside profile. No hidden lines needed on either half unless essential for dimensioning.
* 3. Offset Section: Cutting plane bends at 90-degree right angles to pass through non-collinear features (e.g., passing through a central hub, stepping down to pass through an offset mounting hole). The 90-degree transition offsets are NOT shown with lines in the sectional view.
* 4. Revolved Section: Cross-section of a feature (bar, spoke, rib) rotated 90 degrees in place onto the longitudinal view.
* 5. Removed Section: Cross-section displaced away from the view and placed nearby with clear identification callout (e.g., Section B-B).
* 6. Broken-Out / Local Section: Small partial section bounded by an irregular freehand line to expose an isolated interior detail.

>> 4.4 AutoCAD Editing and Hatching Commands

* STRETCH (S): Select vertices with crossing window (green window); moves vertices while maintaining connectivity of attached geometry.
* EXPLODE (X): Breaks compound objects (Polylines, Blocks, Dimensions, Regions) into basic independent constituent lines and arcs.
* OFFSET (O): Creates parallel concentric curves or parallel lines at a designated offset distance or through point.
* JOIN (J): Merges collinear line segments, connected polyline segments, or continuous spline curves into a single entity.
* REGION (REG): Converts an enclosed 2D boundary loop into a 2D planar solid surface with geometric and physical properties.
* BREAK (BR): Breaks a line or arc between two selected points, creating a gap or dividing into two entities.
* HATCH (H): Fills enclosed boundaries with pattern (ANSI31 standard 45 deg lines, Solid, User-defined). Properties: Pattern, Scale, Angle, Associativity.
* HATCHEDIT (HE): Modifies pattern type, density scale, or island detection of existing hatch objects.

[PROBLEM 1: Full Sectional Front View of a Flanged Bushing]

Problem Statement: A stepped flanged bushing has total length 80 mm. Flange diameter 90 mm, thickness 15 mm. Body diameter 55 mm, length 65 mm. A central cylindrical bore of

diameter 30 mm extends through its entire length. A counterbore of diameter 45 mm and depth 10 mm is cut from the flange end. Draw a Full Sectional Elevation (Section A-A) and Left Side View.

Step-by-Step Construction & Hatching Execution:

* 1. Setup Centerline: Draw horizontal chain-thin centerline.
* 2. Outer Profile Outline: Draw symmetric stepped profile: Flange dia 90 mm (height +45 to -45 from center) for length 15 mm; Body dia 55 mm (+27.5 to -27.5) for length 65 mm. Total length = 80 mm.
* 3. Interior Bores: Bore dia 30 mm (+15 to -15) running through entire length. Counterbore dia 45 mm (+22.5 to -22.5) extending 10 mm into flange.
* 4. Section Slicing & Hatching:
- Solid metal exists between dia 30 mm and dia 55 mm, and between dia 45 mm and dia 90 mm.
- Apply thin continuous hatching lines at 45 degrees across both upper and lower sliced metal sections.
- The bore (air cavity) remains completely clean and unhatched.
* 5. Left Side View: Shows concentric circles of diameters 30 mm, 45 mm (hidden dashed circle), 55 mm, and 90 mm.

[PROBLEM 2: Half Sectional Elevation of Cast Iron Belt Pulley]

Problem Statement: A symmetrical cast iron V-belt pulley has outside rim diameter 160 mm, rim width 40 mm, rim thickness 12 mm. Hub diameter 50 mm, hub length 50 mm, central shaft bore 25 mm with 6 mm x 3 mm keyway. A central solid web plate of thickness 14 mm connects the hub and rim. Draw a Half-Sectional Front View.

Solution & Drafting Conventions:

* 1. Axis & Split: Horizontal centerline represents pulley shaft axis. Vertical centerline divides left exterior view and right sectioned view (or upper section, lower exterior).
* 2. Upper Half in Section: Cutting plane slices through rim, web, and hub. Hub section (between r=12.5 and r=25 mm), web section (thickness 14 mm), and rim section (width 40, thickness 12) are fully hatched at 45 degrees.
* 3. Lower Half as Exterior View: Shows outer face of hub (dia 50), outer rim profile (dia 160), and web surface without any section lines or hidden lines.
* 4. Centerline Boundary: Symmetrical division marked by continuous chain-thin centerline. No solid line drawn along the dividing centerline.

[PROBLEM 3: Offset Section of a Gearbox Cover Plate]

Problem Statement: A gearbox cover plate (200 mm x 120 mm x 20 mm) has a central shaft bearing hole (dia 50 mm) and two mounting bolt holes (dia 14 mm) located offset from the centerline. Illustrate the path of the cutting plane and resulting Offset Section A-B-C-D.

Engineering Solution:

* 1. Cutting Plane Line on Plan View: Enters left edge along line of sight, steps forward by 30 mm at a 90 deg corner to pass through the first bolt hole, steps back along the center axis to slice through the central bearing hole, and steps out through the second bolt hole.
* 2. Sectional View Representation: Drawn as a continuous uninterrupted flat sectional view. The right-angled step transitions are completely invisible in the sectioned view.
* 3. Both the bolt holes and central shaft hole appear as full open cavities with hatched metal around them.

[PROBLEM 4: AutoCAD Hatching and Region Creation Tutorial]

Problem Context: Create associative hatch patterns and modify hatch density for a cross-sectional engine piston drawing.

AutoCAD Command Protocol:

* 1. BOUNDARY / REGION: Type REGION -> Select all boundary lines of the section cut -> Press Enter. Verify command line confirms: "1 loop extracted, 1 Region created".
* 2. HATCH -> Type H -> Select ANSI31 (standard 45 deg iron hatch) -> Set Scale: 1.5 -> Select Pick Points inside region.
* 3. Verify Associativity: Move boundary vertex with STRETCH command; observe that the Hatch pattern automatically dynamically updates and conforms to the new boundary.
* 4. HATCHEDIT: Type HE -> Select hatch -> Change Angle to 135 degrees for adjacent mating sleeve; change scale to 2.0 to differentiate parts.